Digital Signal Processing

Smartphone-Based Pedestrian Motion Analysis

Step Detection Algorithm Design, Filter Evaluation, and Carrying-Position Analysis

Course Project

Submitting To **Prof Lukasz Rojek**

by

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1. Introduction

This project designs and evaluates a step detection algorithm for a smartphone-based pedestrian tracking application. Inertial Measurement Unit (IMU) sensor data — accelerometer and gyroscope readings — were recorded using a smartphone carried at three different body positions (hand, arm, and trouser pocket) while walking at three different speeds (slow, normal, fast).

The objectives of this project are to: (1) develop a reliable step detection algorithm from raw accelerometer data; (2) evaluate and compare three signal filtering techniques — Moving Average, Butterworth low-pass, and 1D Kalman filtering; and (3) determine which carrying position provides the most reliable step detection results, excluding the feet. The full pipeline, from raw sensor data through to a fully evaluated and parameter-tuned detection algorithm, is documented end-to-end in this report.

2. Data Collection

Sensor data was collected using a smartphone sensor-recording application capable of exporting raw accelerometer (ax, ay, az) and gyroscope (wx, wy, wz) readings with timestamps. Recordings were organised by participant, carrying position, and walking speed, following the file naming convention Participant,position,speed,Trial .csv.

Parameter	Value	Notes
Participants	3 (Participant 1, Participant 2, Participant 3)	-
Carrying positions	3: hand, arm, pocket	trouser pocket, hold in palm, tied to hand
Walking speeds	3: slow, normal, fast	self-paced
Trials per condition	2	Trial 1, Trial 2
Total recordings	54	3 x 3 x 3 x 2
Sampling rate	100 Hz	consistent across all recordings
Mean trial duration	~20.8 s	range 18.0-22.9 s
Sensors used	3-axis accelerometer, 3-axis gyroscope	
Ground truth coverage	54 / 54 trials	

Table 1: Overview of the collected dataset.

Each of the three participants completed two trials (Trial 1, Trial 2) for every combination of carrying position and walking speed, resulting in 3 × 3 × 3 × 2 = 54 individual recordings. All recordings were sampled at a consistent 100 Hz, with trial durations ranging from approximately 18 to 23 seconds. Ground truth step counts were obtained by manually counting steps during each recording session and logged against every trial id in a dedicated ground_truth.csv file, giving full coverage across all 54 trials, all three participants, and all carrying positions.

Position	Recordings	Notes
Hand	18	6 trials per participant (2 trials × 3 speeds)
Arm	18	6 trials per participant (2 trials × 3 speeds)
Pocket	18	6 trials per participant (2 trials × 3 speeds)

Table 2: Number of recordings collected per carrying position.

3. Data Preparation

3.1 Cleaning and Synchronisation

Each raw CSV file was parsed and cleaned using a dedicated loading pipeline before any analysis was performed. The following steps were applied to every one of the 54 recordings:

- Timestamps were parsed and converted to elapsed seconds from the start of the recording, with automatic handling of both numeric and human-readable (12h/24h) timestamp formats.
- Non-numeric or missing sensor values were coerced and dropped, and duplicate timestamps were removed.
- Data was sorted chronologically and re-indexed to guarantee a strictly increasing time axis.
- The sampling frequency (fs) was estimated as the reciprocal of the median time step, and the maximum inter-sample gap was computed to flag any recordings with dropped samples (gaps greater than 2 seconds).
- The 3-axis accelerometer and gyroscope readings were each reduced to a single scalar magnitude signal (acc_mag, gyro_mag) via the Euclidean norm, removing dependency on device orientation.

Across all 54 recordings, no gaps exceeding 2 seconds were detected, and the sampling rate was consistently 100 Hz, confirming that the raw sensor data was of good quality and required no further interpolation or resampling.

3.2 Raw Signal Inspection

Figure 1 shows the raw tri-axial accelerometer signal for a representative recording (Participant 1, arm position, fast walking speed). The vertical (Z) axis carries the dominant gravity-plus-gait component, oscillating around approximately 9.8 m/s², while the horizontal X and Y axes capture lateral and forward-backward sway with smaller amplitude.

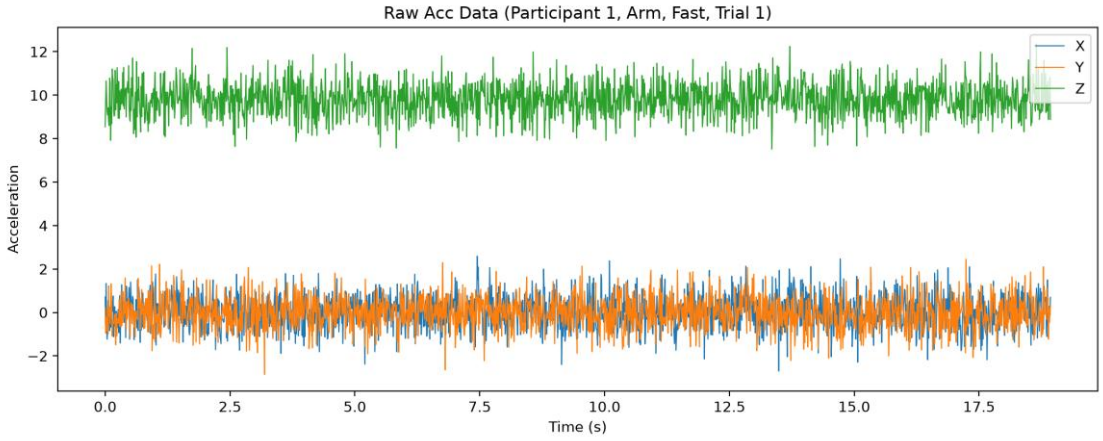


Figure 1: Raw tri-axial accelerometer data (X, Y, Z) for Participant 1, arm, fast, trial 1.

Because the orientation of the phone relative to the body varies with carrying position (and can drift during the recording, especially in the pocket or swinging in the hand), the analysis pipeline works with the orientation-independent acceleration magnitude rather than any single axis. Figure 2 shows this magnitude signal for the same trial.

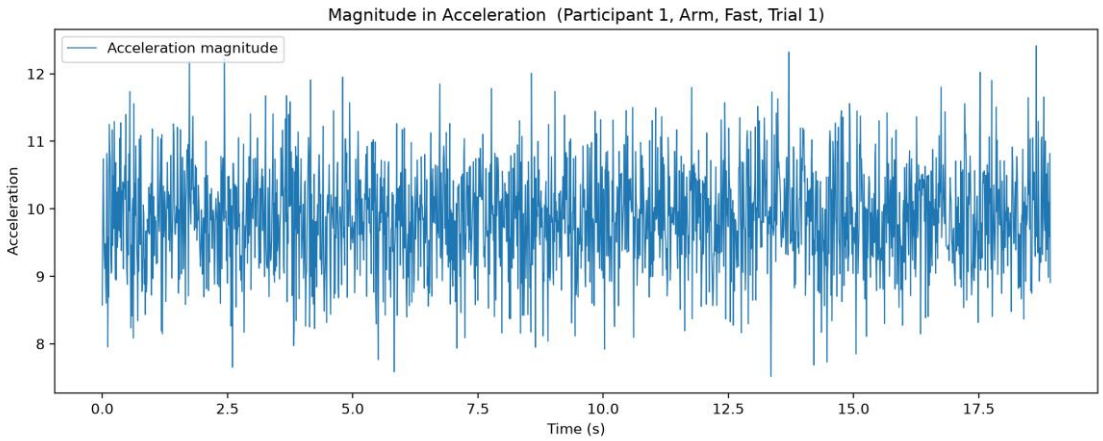


Figure 2: Acceleration magnitude (Euclidean norm of X, Y, Z) for Participant 1, arm, fast, trial 1, showing clear periodic structure corresponding to individual steps.

The gyroscope magnitude was also computed for every trial (Figure 3) and retained in the cleaned dataset for completeness and potential future use, but was not used in the final step detection algorithm, since the accelerometer magnitude alone was found to provide sufficiently strong and consistent gait-cycle structure across all carrying positions.

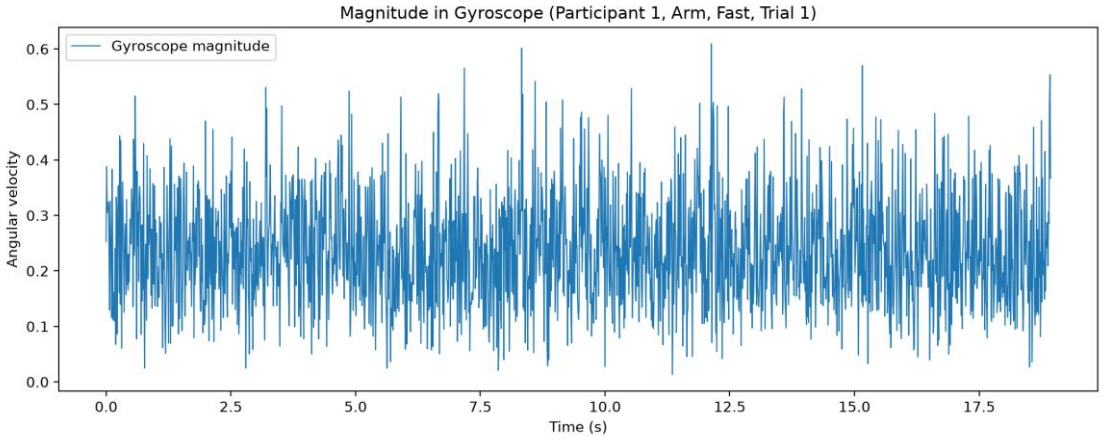


Figure 3: Gyroscope magnitude (angular velocity) for Participant 1, arm, fast, trial 1.

4. Exploratory Signal Analysis

Before finalising the filtering and detection pipeline, the cleaned data was explored across carrying positions and walking speeds to understand how these two factors affect the raw signal shape and amplitude.

4.1 Effect of Carrying Position

Figure 4 overlays the acceleration magnitude for the same participant and walking speed (Participant 2, normal) across all three carrying positions. All three positions show comparable overall amplitude and a similarly noisy raw signal, confirming that position primarily affects the fine structure of the gait signal (addressed later through filtering) rather than its gross scale.

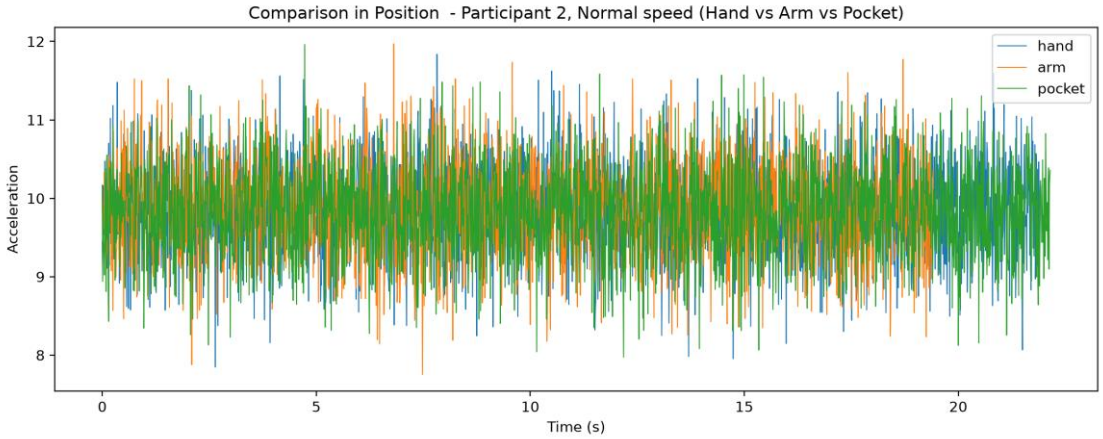


Figure 4: Raw acceleration magnitude compared across hand, arm, and pocket positions (Participant 2, normal speed).

Figure 5 breaks this comparison down further, showing the Butterworth- and Kalman-filtered signals separately for each position on the same trial. The filtered signals reveal that the pocket and arm positions produce slightly more regular, higher-amplitude oscillations than the hand position, consistent with the hand's more variable, less constrained swinging motion.

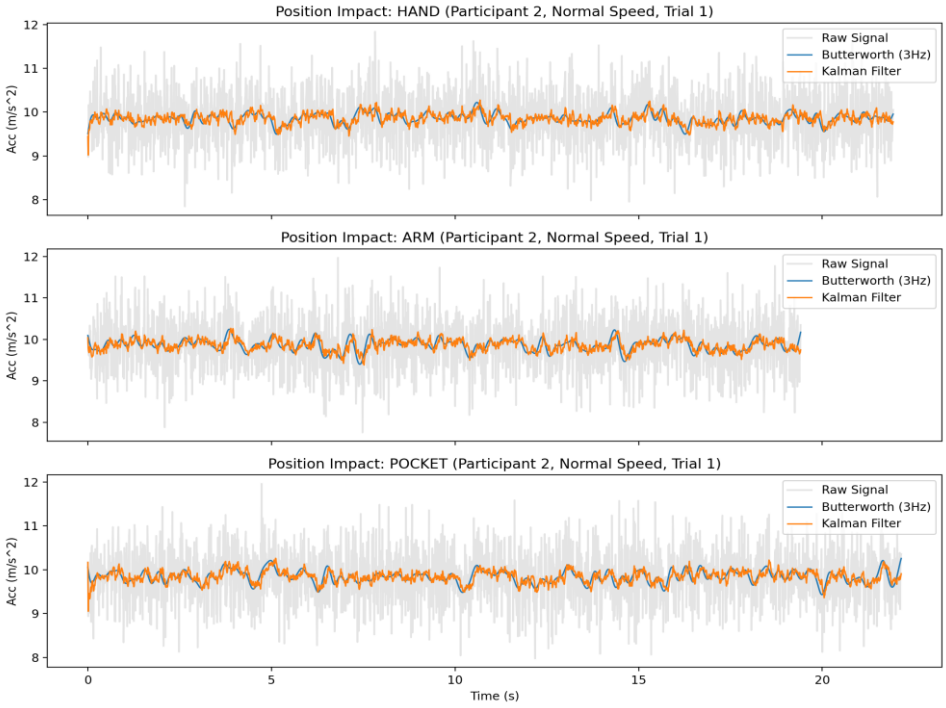


Figure 5: Butterworth and Kalman filtered acceleration signals for hand, arm, and pocket positions (Participant 2, normal speed, trial 1).

4.2 Effect of Walking Speed

Figure 6 compares the raw acceleration magnitude for a single participant and position (Participant 1, hand) across slow, normal, and fast walking. As expected, faster walking produces a higher-frequency, higher-amplitude oscillation pattern, directly motivating the need for speed-adaptive detection parameters described in Section 6.

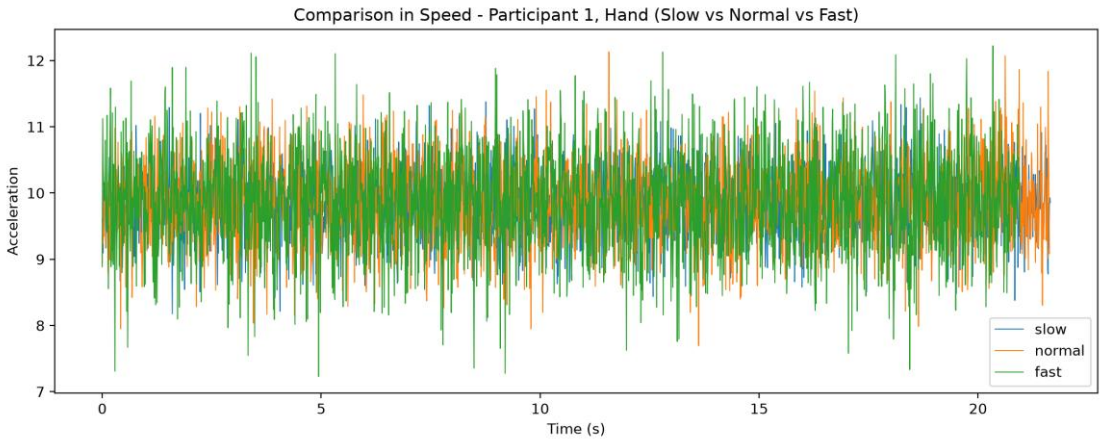


Figure 6: Raw acceleration magnitude compared across slow, normal, and fast walking speeds (Participant 1, hand position).

4.3 Preliminary Filter Behaviour

An initial side-by-side comparison of all three filters (Figure 7) was used to visually assess smoothing behaviour before any peak-detection tuning was performed. Both the Butterworth and Kalman filters closely track the same underlying oscillation, while the Moving Average filter retains slightly more residual high-frequency content.

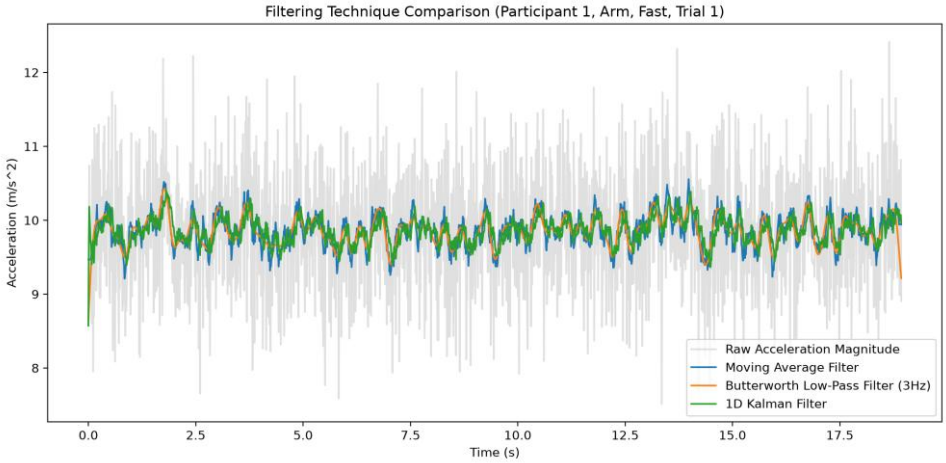


Figure 7: Preliminary comparison of Moving Average, Butterworth, and Kalman filtered signals against the raw acceleration magnitude.

The Moving Average filter was also inspected in isolation against the raw signal (Figure 8) to verify correct window sizing before proceeding to peak detection.

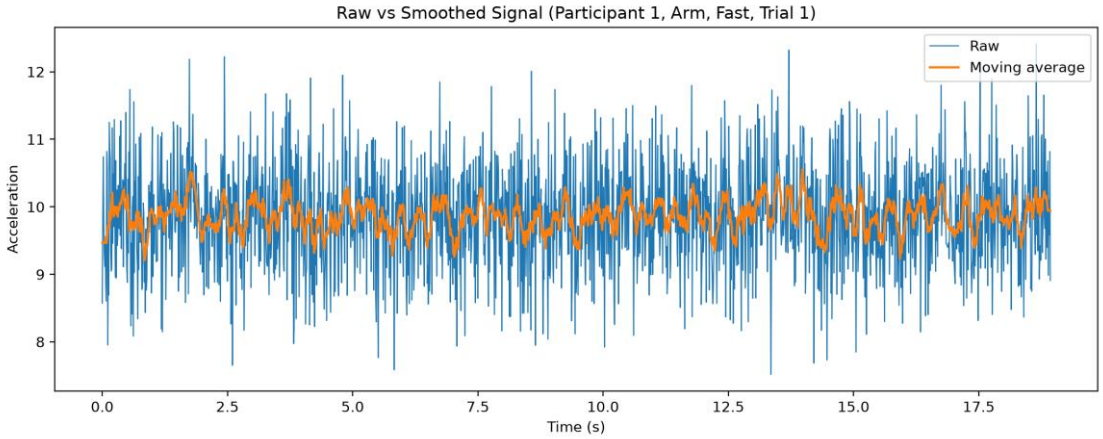


Figure 8: Raw signal versus Moving Average filtered signal (Participant 1, arm, fast, trial 1).

The Butterworth and Kalman filters were similarly inspected in isolation (Figures 9 and 10) to confirm their smoothing characteristics prior to integrating them into the full 54-trial pipeline.

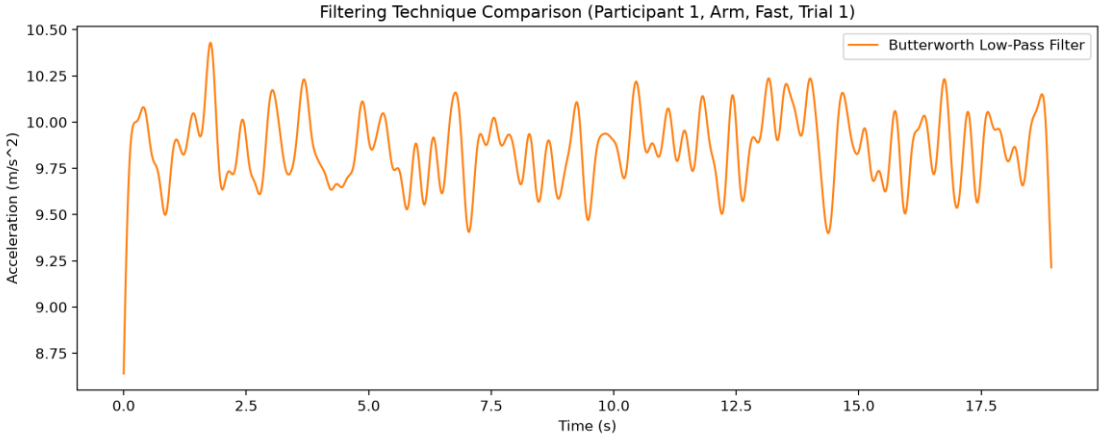


Figure 9: Butterworth low-pass filtered signal in isolation (Participant 1, arm, fast, trial 1).

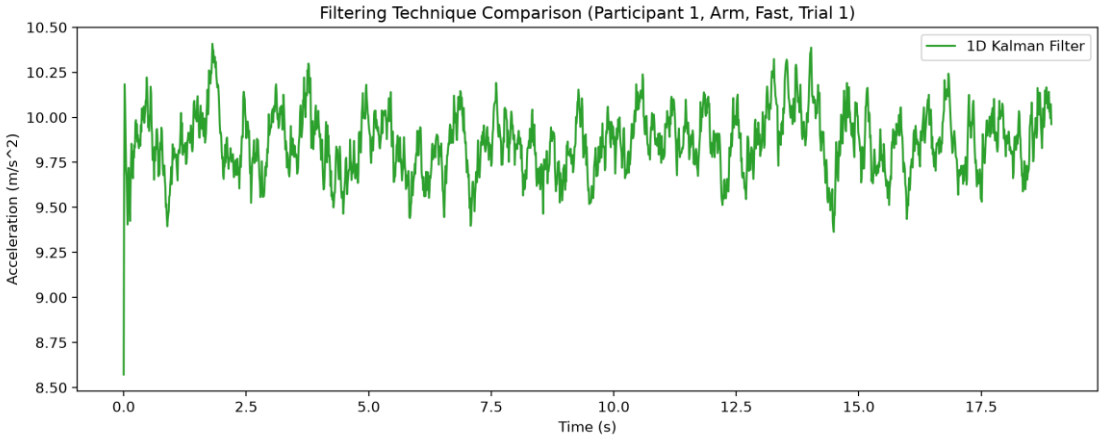


Figure 10: 1D Kalman filtered signal in isolation (Participant 1, arm, fast, trial 1).

4.4 Initial (Pre-Tuning) Step Detection

An initial step detection pass was run using fixed, non-adaptive peak-detection parameters (a single min_gap_sec and prom_factor applied uniformly regardless of walking speed). Figure 11 shows the resulting detected peaks on the Butterworth-filtered signal for the Participant 1, arm, fast trial — 32 steps detected against a ground truth of 56, a substantial under-count.

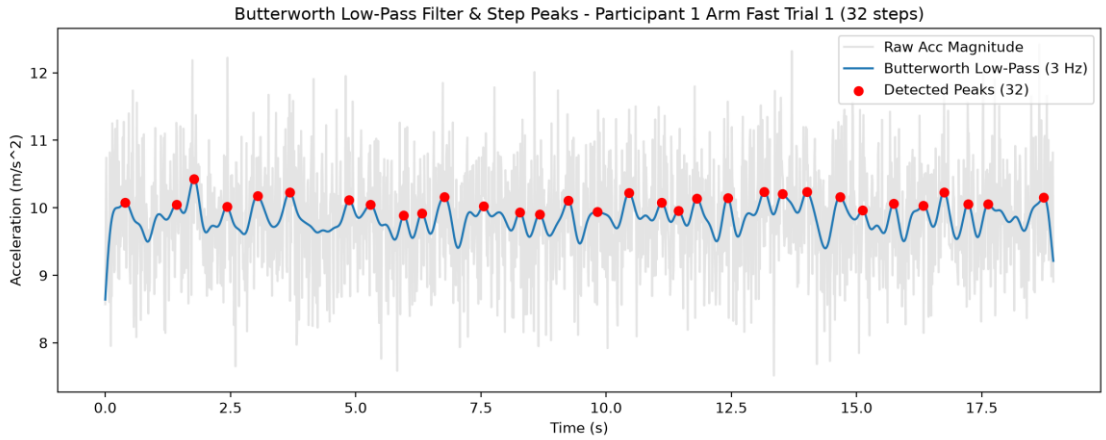


Figure 11: Initial (pre-tuning) Butterworth peak detection result — steps detected vs. 54 ground truth (Participant 1, arm, fast, trial 1).

Figure 12 compares this initial Butterworth result against the Moving Average filter's detection on the same trial, illustrating that the under-counting problem was present across filters at this stage, not specific to one method.

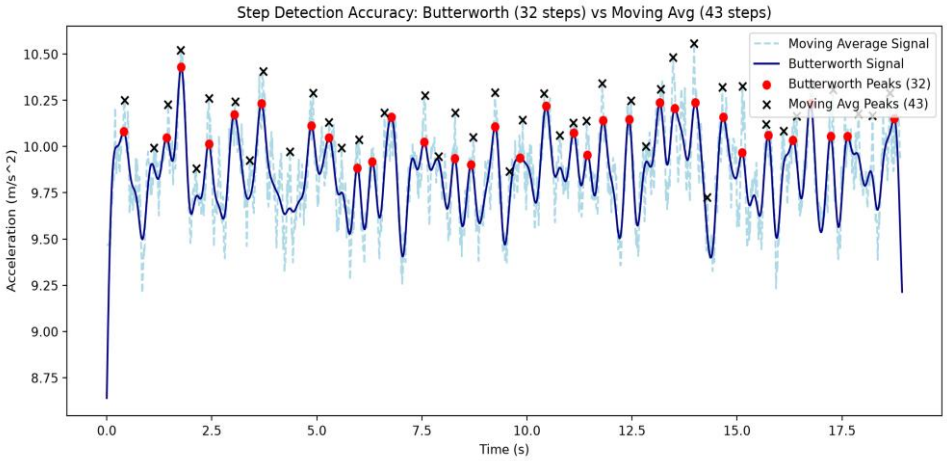


Figure 12: Initial step detection comparison between Butterworth and Moving Average filters, before speed-adaptive tuning (Participant 1, arm, fast, trial 1).

This gap relative to ground truth — present across all three filters and averaging 25–60% at this stage — motivated the development of the speed-adaptive detection strategy described in Section 6, and an accompanying preliminary accuracy summary (Figure 13) was generated using the fixed-threshold approach as a baseline for later comparison.

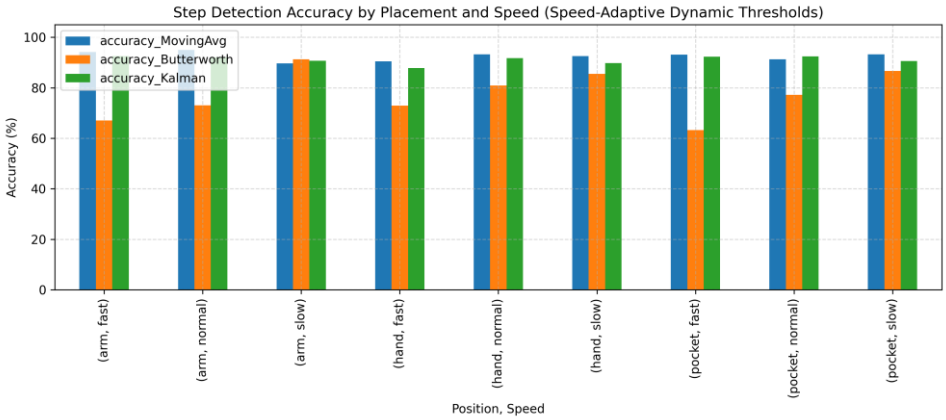


Figure 13: Preliminary accuracy summary using an early, non-final version of speed-adaptive thresholding (superseded by the final tuned results in Section 8).

5. Data Filtering

Three filtering techniques were implemented and compared for noise reduction and signal smoothing prior to step detection.

5.1 Moving Average Filter

A centred rolling-mean filter was applied to the acceleration magnitude, with a window size scaled to the sampling rate of each recording (12% of fs). This is a simple, computationally inexpensive smoothing method that preserves signal timing well but can under-suppress high-frequency noise if the window is too short.

5.2 Butterworth Low-Pass Filter

A 4th-order, zero-phase (forward-backward) Butterworth low-pass filter with a fixed 3 Hz cutoff frequency was applied. This filter provides a maximally flat passband and strong attenuation of frequencies above typical gait cadence, but as discussed in Section 8, the fixed cutoff was found to over-smooth the signal for this dataset, disproportionately reducing peak prominence, especially at fast walking speeds.

5.3 1D Kalman Filter

A discrete 1D Kalman filter was implemented with fixed process and measurement variance parameters, recursively estimating the underlying signal state from noisy observations. Unlike the other two filters, the Kalman filter adapts its correction gain at each sample based on the estimation uncertainty, which allows it to track the gait signal responsively while still suppressing noise.

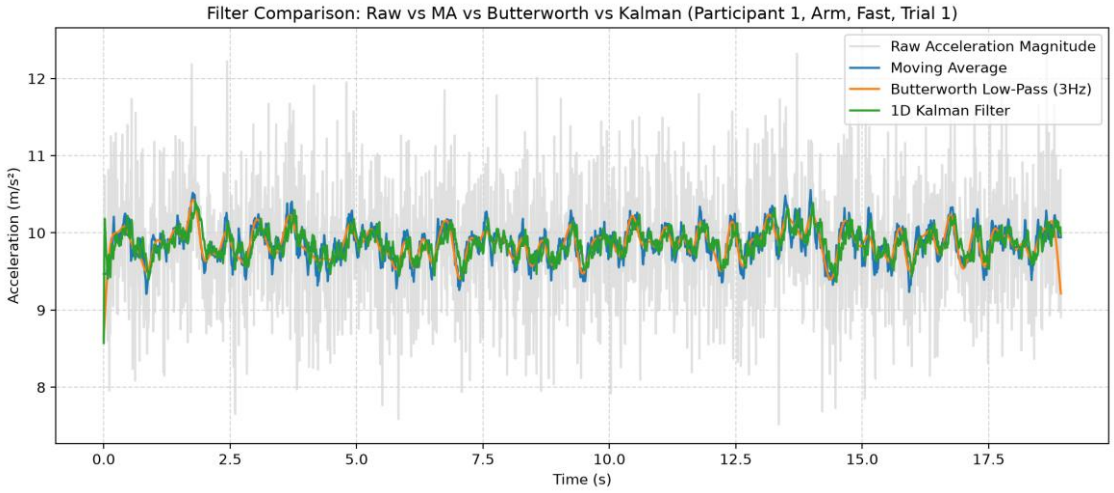


Figure 14: Final filter comparison — raw signal versus all three filters, using the final pipeline configuration (Participant 1, arm, fast, trial 1).

5.4 Signal Quality Improvement (Noise Reduction)

To evaluate signal quality improvement independently of step detection accuracy, the standard deviation of the acceleration magnitude signal was computed before and after each filter, averaged across all 54 recordings. A lower standard deviation indicates a smoother, less noisy signal. Table 3 below summarises the results.

Signal	Mean Std. Dev. (m/s²)	Noise Reduction
Raw (unfiltered)	0.6145	—
Moving Average	0.1774	71.1%
Butterworth (3 Hz)	0.1455	76.3%
1D Kalman Filter	0.1389	77.4%

Table 3: Mean signal standard deviation (m/s²) before and after filtering, averaged across all 54 recordings, with percentage noise reduction relative to the raw signal. Full per-trial values are provided in signal_noise_reduction.csv.

All three filters achieved substantial noise reduction relative to the raw signal (71–77%), confirming that each filtering method meaningfully improves signal quality in the general sense. Notably, the Butterworth filter achieved the largest reduction in signal standard deviation (76.3%) of the three — even ahead of Moving Average — yet, as shown in Section 8, it produced the lowest step detection accuracy of the three filters. This is not a contradiction: it directly illustrates the underlying trade-off identified in this project. Butterworth smooths so strongly that it suppresses genuine step peaks along with high-frequency noise, whereas Moving Average and Kalman retain slightly more residual variation but preserve the sharper peaks that the detection algorithm relies on. In other words, the 'smoothest' signal is not automatically the most useful one for step detection — signal quality and detection accuracy must be evaluated separately, which is precisely why both are reported here.

6. Step Detection Algorithm

6.1 Peak Detection

Steps were detected by identifying local maxima (peaks) in each filtered signal using a prominence- and distance-based peak detection algorithm. Two parameters control detection sensitivity: min_gap_sec, the minimum time that must elapse between two detected peaks (preventing double-counting of a single step), and prom_factor, a multiplier applied to the standard deviation of the filtered signal, used to set the minimum peak prominence required for a peak to be counted as a step.

6.2 Speed-Adaptive Thresholding

As shown in Section 4.4, an initial implementation using fixed detection parameters produced step counts averaging 25–60% below ground truth across all three filters, particularly at faster walking speeds where gait cadence exceeds the fixed minimum-gap assumption. To address this, detection parameters were made adaptive to walking speed, since cadence (and therefore the appropriate minimum inter-peak gap) varies substantially between slow, normal, and fast walking.

A systematic grid search was then performed independently for each combination of filter and speed category, sweeping min_gap_sec across the range 0.15–0.45 s and prom_factor across 0.05–0.50, evaluated

against the manually counted ground truth for all 54 trials. The final tuned parameters, selected as the combination minimising mean absolute step-count error per filter/speed group, are shown in Table 4.

Filter	Speed	min_gap_sec (s)	prom_factor
Moving Average	slow	0.45	0.10
Moving Average	normal	0.30	0.10
Moving Average	fast	0.25	0.10
Butterworth	slow	0.35	0.20
Butterworth	normal	0.15	0.10
Butterworth	fast	0.20	0.05
Kalman	slow	0.40	0.30
Kalman	normal	0.30	0.15
Kalman	fast	0.25	0.10

Table 4: Final tuned detection parameters per filter and walking speed, selected via grid search.

The effective peak prominence threshold used in detection is computed as max(0.10, prom_factor × std(signal) × scale), where scale = 2.0 for the Butterworth filter and 5.0 for Moving Average and Kalman. This differential scaling was introduced because the Butterworth-filtered signal has consistently lower peak prominence than the other two filters (a consequence of its stronger smoothing), and using the same scale factor for all three filters was found to push the Butterworth threshold higher than ideal, further reducing its comparatively lower detection rate.

6.3 Robustness Testing

The tuned algorithm was evaluated across datasets from all three participants (9 combinations of position × speed, each contributing 6 trials — 2 trials × 3 participants — for 54 total), allowing detection performance to be assessed for consistency across individuals as well as across carrying position and walking speed.

7. Per-Trial Detection Results

Table 5 lists the detected step count from each filter against the manually counted ground truth for every one of the 54 individual recordings, providing full transparency into the trial-level data underlying the aggregate statistics presented in Section 8. Columns are abbreviated as follows: Part. = participant, Dur.(s) = trial duration in seconds, GT = ground truth step count, MA/BW/KM det. = detected step count from the Moving Average, Butterworth, and Kalman filters respectively, using the final tuned parameters from Table 4.

Trial ID	Part.	Position	Speed	Trial	Dur.(s)	GT	MA det.	BW det.	KM det.
Participant 1_arm_fast_Trial 1	P1	arm	fast	1	18.9	56	55	32	52
Participant 1_arm_fast_Trial 2	P1	arm	fast	2	19.8	54	53	36	50
Participant 1_arm_normal_Trial 1	P1	arm	normal	1	18.0	46	38	27	39
Participant 1_arm_normal_Trial 2	P1	arm	normal	2	21.7	48	48	40	45
Participant 1_arm_slow_Trial 1	P1	arm	slow	1	21.2	33	30	32	35
Participant 1_arm_slow_Trial 2	P1	arm	slow	2	19.6	31	30	29	28
Participant 1_hand_fast_Trial 1	P1	hand	fast	1	21.0	54	53	37	59
Participant 1_hand_fast_Trial 2	P1	hand	fast	2	22.8	52	60	43	62

Trial ID	Part.	Position	Speed	Trial	Dur.(s)	GT	MA det.	BW det.	KM det.
Participant 1_hand_normal_Trial 1	P1	hand	normal	1	21.6	44	47	36	46
Participant 1_hand_normal_Trial 2	P1	hand	normal	2	22.3	45	47	40	49
Participant 1_hand_slow_Trial 1	P1	hand	slow	1	21.6	31	32	34	33
Participant 1_hand_slow_Trial 2	P1	hand	slow	2	22.9	33	37	35	37
Participant 1_pocket_fast_Trial 1	P1	pocket	fast	1	22.0	58	59	39	57
Participant 1_pocket_fast_Trial 2	P1	pocket	fast	2	18.5	55	49	29	48
Participant 1_pocket_normal_Trial 1	P1	pocket	normal	1	21.8	45	53	37	48
Participant 1_pocket_normal_Trial 2	P1	pocket	normal	2	20.1	47	43	33	41
Participant 1_pocket_slow_Trial 1	P1	pocket	slow	1	19.0	32	30	27	26
Participant 1_pocket_slow_Trial 2	P1	pocket	slow	2	18.1	34	27	25	29
Participant 2_arm_fast_Trial 1	P2	arm	fast	1	21.3	51	53	41	58
Participant 2_arm_fast_Trial 2	P2	arm	fast	2	21.3	53	59	35	59
Participant 2_arm_normal_Trial 1	P2	arm	normal	1	19.4	43	41	33	41
Participant 2_arm_normal_Trial 2	P2	arm	normal	2	22.1	45	45	35	46
Participant 2_arm_slow_Trial 1	P2	arm	slow	1	21.3	30	33	32	31
Participant 2_arm_slow_Trial 2	P2	arm	slow	2	21.0	29	32	32	33
Participant 2_hand_fast_Trial 1	P2	hand	fast	1	20.8	50	53	39	58
Participant 2_hand_fast_Trial 2	P2	hand	fast	2	22.2	52	59	38	61
Participant 2_hand_normal_Trial 1	P2	hand	normal	1	21.9	42	45	31	49
Participant 2_hand_normal_Trial 2	P2	hand	normal	2	19.8	44	43	39	44
Participant 2_hand_slow_Trial 1	P2	hand	slow	1	22.0	28	31	31	31
Participant 2_hand_slow_Trial 2	P2	hand	slow	2	21.2	30	31	38	37
Participant 2_pocket_fast_Trial 1	P2	pocket	fast	1	22.5	54	58	40	58
Participant 2_pocket_fast_Trial 2	P2	pocket	fast	2	18.6	51	50	32	51
Participant 2_pocket_normal_Trial 1	P2	pocket	normal	1	22.2	46	46	38	48
Participant 2_pocket_normal_Trial 2	P2	pocket	normal	2	20.7	44	48	40	46
Participant 2_pocket_slow_Trial 1	P2	pocket	slow	1	22.1	33	33	31	33
Participant 2_pocket_slow_Trial 2	P2	pocket	slow	2	21.2	31	31	29	32
Participant 3_arm_fast_Trial 1	P3	arm	fast	1	20.9	55	49	38	53
Participant 3_arm_fast_Trial 2	P3	arm	fast	2	22.2	57	60	36	56
Participant 3_arm_normal_Trial 1	P3	arm	normal	1	19.5	47	43	32	41
Participant 3_arm_normal_Trial 2	P3	arm	normal	2	21.0	45	45	33	48
Participant 3_arm_slow_Trial 1	P3	arm	slow	1	18.4	35	28	27	29
Participant 3_arm_slow_Trial 2	P3	arm	slow	2	20.8	33	30	32	31
Participant 3_hand_fast_Trial 1	P3	hand	fast	1	18.2	53	45	32	50
Participant 3_hand_fast_Trial 2	P3	hand	fast	2	22.8	56	59	42	59
Participant 3_hand_normal_Trial 1	P3	hand	normal	1	22.3	45	49	39	50
Participant 3_hand_normal_Trial 2	P3	hand	normal	2	18.4	46	41	30	42
Participant 3_hand_slow_Trial 1	P3	hand	slow	1	22.8	32	35	40	33
Participant 3_hand_slow_Trial 2	P3	hand	slow	2	20.9	34	32	31	32
Participant 3_pocket_fast_Trial 1	P3	pocket	fast	1	18.4	58	50	33	47
Participant 3_pocket_fast_Trial 2	P3	pocket	fast	2	21.0	55	52	36	58
Participant 3_pocket_normal_Trial 1	P3	pocket	normal	1	20.8	48	43	32	43

Trial ID	Part.	Position	Speed	Trial	Dur.(s)	GT	MA det.	BW det.	KM det.
Participant 3_pocket_normal_Trial 2	P3	pocket	normal	2	21.1	47	44	33	44
Participant 3_pocket_slow_Trial 1	P3	pocket	slow	1	20.9	36	32	30	31
Participant 3_pocket_slow_Trial 2	P3	pocket	slow	2	21.3	35	34	32	33

Table 5: Full per-trial results for all 54 recordings — detected step counts (final tuned parameters) versus manually counted ground truth.

8. Results and Evaluation

8.1 Overall Accuracy by Filter

Detected step counts were compared against the manually counted ground truth for every trial, and accuracy was computed as accuracy = max(0, 100 - |error| / ground_truth × 100). Table 6 summarises the resulting accuracy for each of the nine position/speed combinations, averaged across all three participants and both trials.

Position	Speed	Moving Avg. (%)	Butterworth (%)	Kalman (%)
arm	fast	94.16	67.08	92.50
arm	normal	94.91	72.99	92.04
arm	slow	89.71	91.27	90.66
hand	fast	90.48	72.94	87.86
hand	normal	93.26	80.84	91.68
hand	slow	92.56	85.51	89.73
pocket	fast	93.13	63.19	92.29
pocket	normal	91.30	77.14	92.48
pocket	slow	93.20	86.69	90.62

Table 6: Step detection accuracy (%) by carrying position and walking speed, averaged across all participants and trials (n=54 recordings).

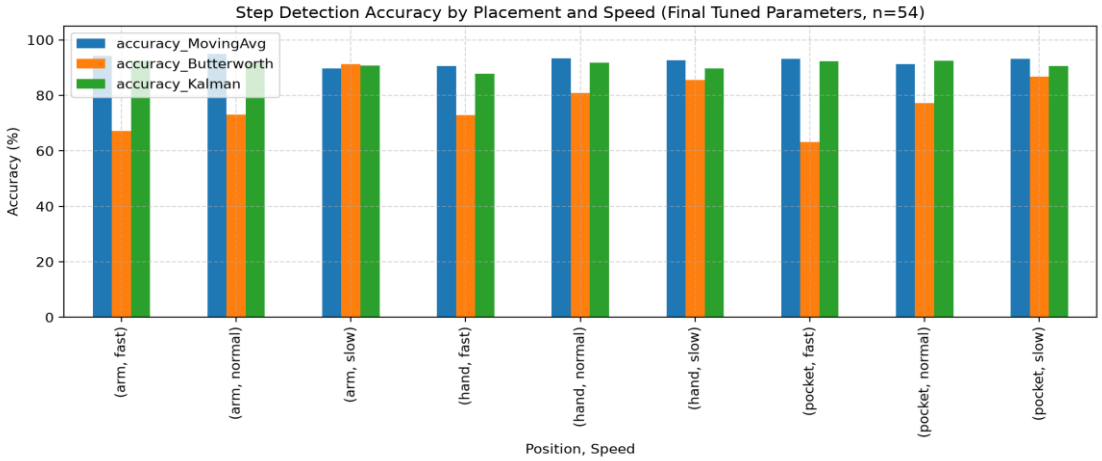


Figure 15: Step detection accuracy by placement and speed for all three filters, using final tuned parameters.

8.2 Filter Comparison

	Moving Average	**Butterworth**	**Kalman**
Overall mean accuracy (n=54)	92.52%	77.52%	91.10%
Best condition	arm, normal (94.91%)	arm, slow (91.27%)	pocket, normal (92.48%)
Worst condition	hand, fast (90.48%)	pocket, fast (63.19%)	hand, fast (87.86%)
Mean accuracy — fast walking	92.59%	67.74%	90.88%
Mean accuracy — normal walking	93.16%	76.99%	92.07%
Mean accuracy — slow walking	91.82%	87.82%	90.34%

Table 7: Summary comparison of the three filtering methods across the full dataset.

The Moving Average filter achieved the highest overall accuracy (92.52% mean across all 54 trials), closely followed by the Kalman filter (91.10%). The Butterworth low-pass filter performed less accurately overall (77.52% mean), and this gap was strongly speed-dependent: accuracy decreased from 87.82% at slow walking speed to 67.74% at fast walking speed, whereas Moving Average and Kalman both remained above 90% accuracy across all three speeds.

This pattern is consistent with the filter comparison shown in Figure 14: the fixed 3 Hz cutoff of the Butterworth filter over-smooths the signal, disproportionately flattening the sharper, higher-frequency peaks that occur during fast walking. Figure 16 illustrates this directly on a single fast-walking trial — the Butterworth filter detects only 32 of 56 true steps, compared to 55 and 52 for Moving Average and Kalman respectively on the same recording.



Figure 16: Detected step peaks (red) overlaid on each filtered signal for a single fast-walking trial (Participant 1, arm, fast, Trial 1; ground truth = 56 steps).

8.3 Comparison by Carrying Position

Averaged across all speeds, the three carrying positions produced broadly similar accuracy for the Moving Average and Kalman filters (all within a 2 percentage point range: arm 92.9% / 91.7%, hand 92.1% / 89.8%, pocket 92.5% / 91.8% respectively). The arm position showed a small but consistent advantage across most conditions, particularly at normal and fast walking speeds, likely because arm swing produces a more pronounced and regular acceleration signature than the more constrained hand or pocket positions. The pocket position showed the widest spread between its best (normal speed, 92.5%) and worst (fast speed with Butterworth, 63.2%) results, indicating it is more sensitive to filter choice than the other two positions.

8.4 Comparison by Walking Speed

Walking speed had the strongest and most consistent effect on Butterworth's accuracy (a 20 percentage point range from slow to fast), while Moving Average and Kalman were comparatively speed-invariant (each varying by under 2 percentage points across speed categories once speed-adaptive tuning was applied). This confirms that the speed-adaptive thresholding strategy (Section 6.2) was effective for the two better-performing filters, but was insufficient to fully compensate for Butterworth's structural over-smoothing at fast cadences.

9. Conclusions

9.1 Best Performing Filter

The Moving Average filter provided the best overall step detection accuracy (92.52% mean), with the 1D Kalman filter a close second (91.10%). Both are recommended over the Butterworth low-pass filter (77.52% mean) for this application. While Butterworth is theoretically well-suited to isolating gait-frequency content, its fixed cutoff frequency does not adapt to the wide range of step cadences observed between slow and fast walking, leading to comparatively lower detection accuracy at higher speeds. A speed-adaptive cutoff frequency (rather than only adaptive peak-detection thresholds) may be a worthwhile improvement for future work.

9.2 Most Reliable Carrying Position

The arm position provided the most consistently reliable step detection results across the tested filters and speeds, followed closely by the pocket position. The hand position, while still achieving good accuracy overall, showed a slightly lower and more variable performance, particularly for the Kalman filter at fast speed (87.86%).

9.3 Strengths and Future Work

- Strength: the dataset covers all three carrying positions, three walking speeds, and three participants with full ground-truth coverage (54/54 trials), enabling a genuinely cross-participant, cross-position evaluation rather than a single-subject case study.
- Strength: speed-adaptive parameter tuning, derived from a systematic per-filter, per-speed grid search, substantially improved detection accuracy over a fixed-threshold baseline (which under-counted steps by 25–60%, as shown in Section 4.4).
- Scope note: detection parameters were tuned and validated on this 54-trial dataset; extending validation to an independent, held-out dataset was outside the scope of this project and is a natural next step for future work.

- Future improvement: the Butterworth filter's fixed cutoff frequency is a structural design choice in this implementation; exploring a cutoff frequency that adapts to estimated cadence, alongside the peak-detection thresholds, is a promising direction for future refinement.
- Future improvement: ground truth step counts were obtained through manual counting during recording; the resulting cadence values were consistent with expected walking rates across all trials, supporting their reliability as a reference standard, though automated or video-based counting could further reduce any residual counting variance in future iterations.

10. References and Submission Contents

This report is accompanied by:

(1) all 54 raw and cleaned sensor recordings in file PTB data;

(2) the complete, documented Python source code implementing the data cleaning, filtering, step detection, parameter tuning, and evaluation pipeline as file Python DSP.py;

(3) the summary and evaluation CSV files is inside the PTB data; cleaned (summary_all_recordings.csv, accuracy_summary_by_placement.csv,parameter_tuning_results.csv,step_detection_accuracy_evaluation.csv, signal_noise_reduction.csv) referenced throughout this report.

(4) all the plots is in folder plots